

COURSE DESCRIPTION			
Program	Chemical Engineering, Food Processing Technology, Polymer Technology, Printing Technology, Textile Technology		
Course Title	Mathematics for Applied Engineering		
Course Code	2001D	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	3:1:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course introduces fundamental mathematical tools essential for engineering studies. It includes the study of complex numbers, vector algebra, statistics and basic integral calculus with differential equations, aimed at supporting analytical thinking and practical problem-solving. The course prepares students to apply mathematical concepts effectively in engineering and technological contexts.

Course Objectives

1	To provide comprehensive introductory-level coverage of complex numbers
2	To familiarize students with the basics of vector algebra
3	To introduce the fundamentals of integral calculus and differential equations for modelling and solving engineering-related problems.
4	To develop an understanding of statistics for data analysis and decision-making in engineering contexts

Course Pre-requisite

Topic	Course code	Course Title	Semester
Trigonometry, Determinants, Differentiation	1002	Fundamentals of Engineering Mathematics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply concepts of complex numbers to solve problems.	Apply
CO2	Identify and apply vector algebra techniques to solve problems involving scalar and vector quantities in engineering contexts.	Apply
CO3	Use integration techniques and differential equations to solve basic engineering problems.	Apply
CO4	Apply statistical methods to analyse and interpret data related to engineering and real-life situations.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO1	3	-	-	-	-	-	-	-	-	-	-
CO1	3	-	-	-	-	-	-	-	-	-	-
CO1	3	-	-	-	-	-	-	-	-	-	-
CO1	12	-	-	-	-	-	-	-	-	-	-
CO1	3	-	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	1	0	0	5.5	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Complex Numbers	The Complex Number System, Fundamental operations with complex numbers, Modulus and Amplitude, Graphical representation of complex numbers, Polar form	11	4
Module II	Vector Algebra	Scalars and vectors, vector addition, subtraction, multiplication by a scalar, Dot product, Cross product, Applications	12	3
Module III	Integral Calculus and Differential Equations	Integration as the reverse process of differentiation, Standard results, Integration by parts, Integration by substitution of integral of $f(ax+b)$, $f(x) \cdot f'(x)$ and $f'(x)/f(x)$, Definite integrals, Differential Equations	11	4
Module IV	Statistics	Data collection and classification, Frequency distribution and graphical representation, Measures of central tendency, Measures of dispersion,	11	4

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
The complex number system	Understand the complex number system and identify the real part and imaginary part	L	CO1	Understand	1
Conjugate of a complex number, Equality of complex numbers	Determine the conjugate of a complex number, Test the equality of complex numbers by comparing their real and imaginary parts.	L	CO1	Apply	1
Fundamental operations with complex numbers : Addition, Subtraction, Multiplication, Division	Perform fundamental operations on complex numbers	L	CO1	Understand	5
Graphical Representation	Represent complex numbers graphically on the Argand (complex) plane.	L	CO1	Understand	1
Absolute value and Amplitude of a Complex Number	Determine the Modulus (Absolute value) and amplitude (argument) of a complex number in degrees	L	CO1	Apply	2
Polar Form	Express a complex number in polar form	L	CO1	Understand	1
Problems	Solve simple engineering problems involving complex numbers.	T	CO1	Apply	4
Module II					
Concept of scalars and vectors, Types of vectors, The unit vectors i, j, k , Position vector of a point, Representation of a vector in terms of i, j, k	Distinguish between scalar and vector quantities and identify components of a vector	L	CO2	Understand	1
Magnitude of a vector	Determine magnitude of a vector and find the unit vector in the direction of a given vector	L	CO2	Understand	1
Addition and subtraction of vectors, Multiplication by a scalar	Perform addition and subtraction of vectors, multiplication of vectors by a scalar	L	CO2	Understand	2
Dot Product	Compute the dot product of vectors and apply it to determine angle between vectors and perpendicularity	L	CO2	Apply	2
Cross Product	Evaluate the cross product of vectors, parallelism	L	CO2	Apply	2
Application of vectors in Engineering : area of parallelogram, area of triangle, work done, moment	Apply vector algebra to solve engineering problems involving area of a parallelogram, area of a triangle, work done, and moment of a force	L	CO2	Apply	4

Problems	Solve and conceptual problems based on vector operations	T	CO2	Apply	3
Module III					
Integration as the reverse process of differentiation	Understand integration as the inverse process of differentiation	L	CO3	Remember	1
Integral of standard functions(x^n , e^x , $1/x$, $\sin x$, $\cos x$, $\sec^2 x$, $\operatorname{cosec}^2 x$, $\sec x \tan x$, $\operatorname{cosec} x \cot x$)	Remember standard integrals of basic functions	L	CO3	Remember	1
Rules of integration and simple problems, Integration by substitution of integral of $f(ax+b)$, $f(x) \cdot f'(x)$ and $f'(x)/f(x)$	Use basic rules of integration (constant multiple, sum and difference rules) to solve simple integrals.	L	CO3	Understand	2
Integration by parts rule and problems	Apply the integration by parts method to evaluate integrals of product of two functions	L	CO3	Apply	2
Definite integrals and simple problems	Evaluate definite integrals	L	CO3	Apply	2
Differential equations, order and degree of D.E	Identify and classify differential equations by determining their order and degree.	L	CO3	Understand	1
First order differential equations in variable separable form, First order L.D.E, $dy/dx + Py = Q$	Solve first order differential equations (variable separable and linear)	L	CO3	Apply	2
Problems	Evaluate integrals and solve first order differential equations	T	CO3	Apply	4
Module IV					
Definition and importance of data, Types of data: Primary and Secondary data, Qualitative and Quantitative data. Methods of data collection (survey, observation, records), Raw data and array formation, Classification of data, Introduction to frequency distribution	Define, classify and organise data, introduce frequency distribution.	L	CO4	Remember	2
Frequency distribution, Class intervals, Class limits, Class boundaries, Graphical representation of data : Histogram, Frequency polygon, Ogive (cumulative frequency curve), Interpretation of graphs	Identify class intervals, class limits and class boundaries, represent data graphically using histograms, frequency polygons, and ogives.	L	CO4	Apply	3
Concept and importance of averages, Measures of central tendency : Arithmetic Mean (ungrouped and grouped data), Median, Mode.	Compute arithmetic mean, median, and mode	L	CO4	Apply	3
Concept of dispersion and variability, Measures of dispersion: Mean Deviation, Standard Deviation, Variance.	Compute standard deviation and variance for grouped and ungrouped data sets.	L	CO4	Apply	3
Problems	Solve problems on frequency distribution, measures of central tendency, measures of dispersion	T	CO4	Apply	4

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Find conjugate of complex numbers, Determine modulus and amplitude and write polar form of complex numbers	4
2	Solve problems related to equality, addition and subtraction of complex numbers	4
3	Multiply and divide complex numbers	4
4	Find the applications of complex numbers in respective engineering areas.	3
Module I		
5	Identify scalars and vectors, Solve problems on vector addition and subtraction, Multiply vectors by a scalar	4
6	Perform dot product and cross product of vectors	4
7	Solve problems to find work done and moment	4

8	Find the applications of vectors in respective engineering areas.	3
Module I		
9	Find integrals of functions	4
10	Evaluate definite integrals	4
11	Solve first order differential equations	4
12	Find the applications of integral calculus in respective engineering areas.	3
Module I		
13	Identify the type of data and method of collection using various context, Identify Class intervals, Class limits, Frequencies using the examples, and represent it graphically.	5
14	Calculate Mean, Median, and Mode for the data collected from relevant fields, and prepare a comparison table.	5
15	Calculate Standard Deviation for small data sets. Collect relevant data from the field pertaining to the branch, analyze the data, and interpret the results.	5
Total Self-Learning hours		60

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
	Mathematics for Applied Engineering		SITTTR Kalamassery

REFERENCE			
Sl. No.	Title	Author	Publication
1	Schaum's outline of Complex Variables	Murray R. Spiegel, Seymour Lipschutz, John J. Schiller	Mc Graw Hill
2	Schaum's outline of Vector Analysis	Murray R. Spiegel	Mc Graw Hill
3	Schaum's outline of Statistics	Murray R. Spiegel, Larry Stephens	Mc Graw Hill
4	Advanced Engineering Mathematics, 10th Ed.	Erwin Kreyszig	Wiley & Sons
5	A Textbook of Engineering Mathematics	N.P. Bali, Manish Goyal	University Science Press
6	Advanced Engineering Mathematics	B. S. Grewal	Mercury Learning & Information

WEB RESOURCES		
Sl. No.	URL	Modules Covered
	NIL	

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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